

| STAT248 Lecture 3 Measures of Dependence

| 1 Mean and Variance

| 1.1 White Noise

对于 white noise time series,

Mean:

$$\mu_{wt} = \mathbb{E}[w_t] = 0, \quad \forall t$$

Variance (对于 Gaussian white noise series):

$$\text{Var}[w_t] = 1$$

| 1.2 Moving Average

对于 moving average, 尽管其会引入 correlation structure, 但并不会改变 mean,

Mean:

$$\mu_{vt} = \mathbb{E}[v_t] = \frac{1}{n} \sum_{k=-\frac{n-1}{2}}^{\frac{n-1}{2}} \mathbb{E}[w_{t+k}] = 0$$

Variance:

$$\text{Var}[v_t] = \text{Var} \left[\frac{1}{n} \sum_{k=-\frac{n-1}{2}}^{\frac{n-1}{2}} w_{t+k} \right] = \frac{1}{n^2} \sum_{k=-\frac{n-1}{2}}^{\frac{n-1}{2}} \text{Var}[w_{t+k}] = \frac{\sigma^2}{n}$$

| 1.3 Random Walk

对于 random walk,

Mean:

$$\mathbb{E}[x_t] = \sum_{j=1}^t \mathbb{E}[w_j] = 0$$

Variance:

$$\text{Var}[x_t] = \text{Var} \left[\sum_{j=1}^t w_j \right] = t\sigma^2$$

| 1.4 Random Walk with Drift

对于 random walk with drift δ ,

Mean:

$$\mathbb{E}[x_t] = \delta t + \sum_{j=1}^t \mathbb{E}[w_j] = \delta t$$

Variance:

$$\text{Var}[X_t] = \text{Var} \left[\sum_{j=1}^t w_j \right] = t\sigma^2$$

| 1.5 Signal Plus Noise

对于 signal plus noise time series (以 sinusoid 为例),

Mean:

$$\begin{aligned} \mu_{xt} = \mathbb{E}[x_t] &= \mathbb{E}[A \cos(2\pi t + \phi) + w_t] \\ &= \mathbb{E}[A \cos(2\pi t + \phi)] + \mathbb{E}[w_t] \\ &= \mathbb{E}[A \cos(2\pi t + \phi)] \\ &= A \cos(2\pi t + \phi) \end{aligned}$$